

a)

→ We can charge a body positively by induction ~~using~~ by following a set of steps: First we need to place a conductor we want to charge positively on top of a insulator making sure it doesn't touches the ground or any other conductor. Then we need a negative charged object (either by friction or induction) and place it near the conductor we want to neutral conductor we want to charge. The negative charged on the negatively charged body will attract bound charges (or positive charges) on the neutral conductor and push the negative charges (free charges) on the other side. Then we need to perform earthing to the ~~the~~ (free charges) side of the conductor and doing this will transfer negative charges to the ground or just pull positive electron ~~charges~~ charges.

$$C = \frac{Q}{V}$$

$$C = [\log_{10} N] +$$

charges and neutralize the negative charges.

After that pull the negatively charged body from the conductor and how the neutral conductor is positively charged.

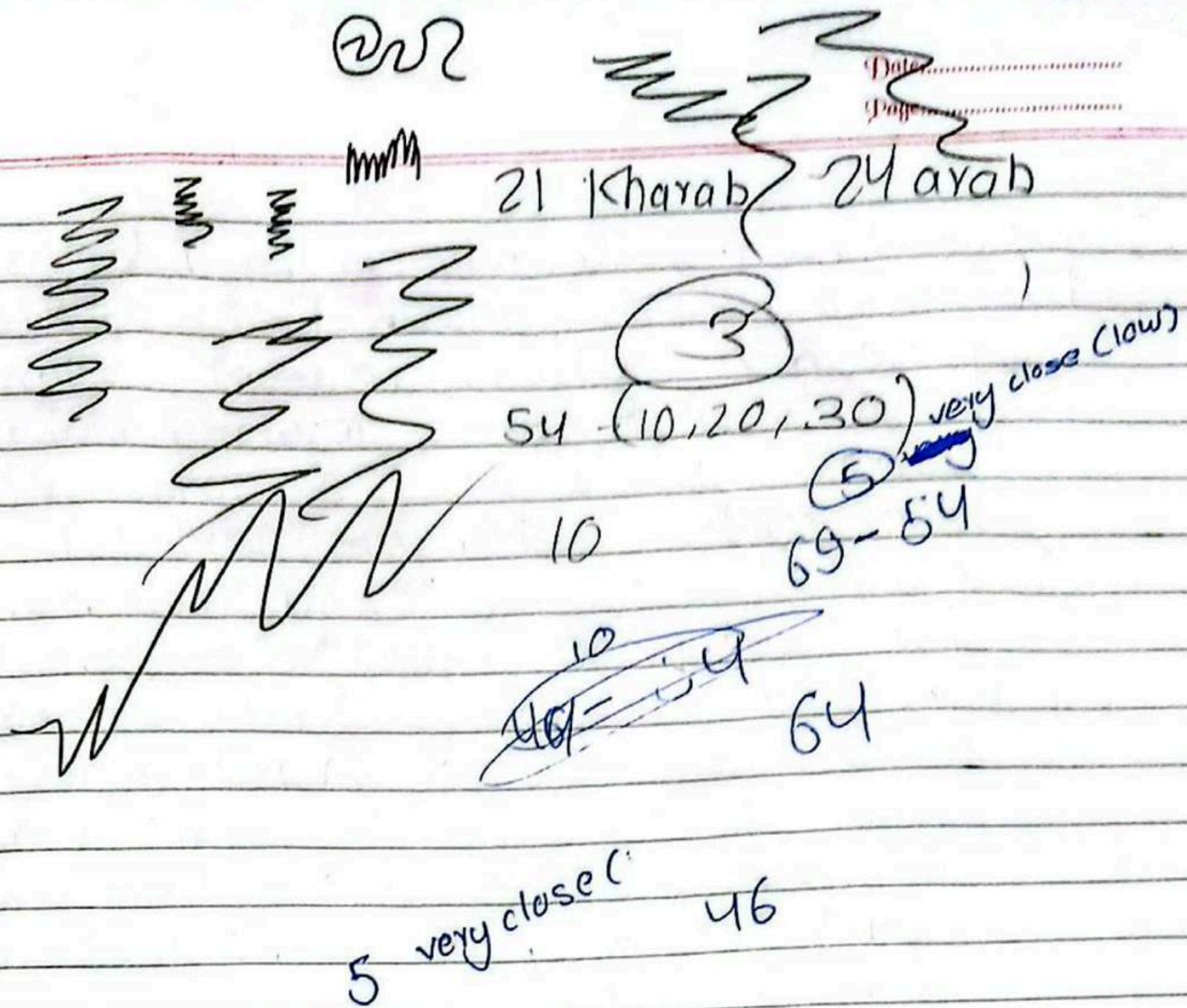
$$2750 \times 3 = 8250$$

$$20750$$

$$201100$$

$$99000$$

1



Action of point is a phenomenon of electric charges in which charges tend to accumulate more in the ~~the~~ area where there is more curve and charges actually leak from pointy surfaces.

We can use this property to our benefit by ~~no~~ placing a sharp pointed metal on top of a building and ground it into earth. This ensure if thunder occurs then instead of hitting other infrastructure or organisms, it will ~~transf~~ tend to transfer itself to the earth through the sharp pointed metal on the top of building.

$$x = \frac{a \sin \theta + b \cos \theta}{a+b}$$

$$\text{O}_a \quad \frac{Q+q}{2}$$

$$G = \frac{F d^2}{m^2}$$

$$\frac{[M L T^{-2}] \times [L^2]}{[M^2]}$$

$$[M L T^{-2}] \cdot [L^2] L^0$$

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$$\rho = \frac{m}{V}$$

$$= \frac{Kg}{m^3} \quad [M L^{-3} T^0]$$

- a) • The statement, "electric charge of a body is quantized", is correct and it says that electric charge is the integral multiple meaning electric charge cannot be in fraction like $+\frac{1}{2}q$, $+\frac{2}{3}q$, and can only be in whole number.

$$-273.15 = 0.15^{\circ}\text{C} \quad C = K - 273$$

e)

-> Yes, charging by friction does follow the conservation of charge because, in the method of friction, the body with the higher work function will be positively charged and lower work function will be negatively charged.

Yes, charging by friction does follow the conservation of charge because, when we take two different ~~the~~ body with different charges, and rub them, the one with lower work function will be ~~neg~~ positively charged and the one with higher work function will be negatively charged. ~~and they will~~

f) Iron chains touching to the ground are suspended from a vehicle carrying inflammable materials, because, in order to protect it from static electricity or lightning bolt with the help of the phenomenon of action of points.

b)

We know, $C = \frac{1}{3} F$ $F = 3C$

Now, $C = F - 32$

$$\frac{100}{3} \cdot F = \frac{180}{180} F - 32$$

$$\frac{C}{100} = \frac{3C - 32}{180}$$

$$\frac{C}{5} = \frac{3C - 32}{9}$$

$$9C = 3C - (150 + 10)$$

$$3C = -160$$

$$C = \frac{-160}{3} = -53.33$$

$$\therefore C = -5.33^\circ C$$

$$\frac{\Delta C}{100} = \frac{\Delta F}{180}$$

$$\frac{\Delta C}{5} = \frac{\Delta F}{9}$$

$$25 = \frac{\Delta F}{9}$$

$$\Delta F = 45$$

d)

Soln,

We know Given:

$$F^{\circ} = K$$

We know,

$$\frac{F-32}{180} = \frac{K-273}{373-273}$$

$$\frac{F-32}{180} = \frac{F-273}{100}$$

$$5F - 160 = 9F - (1800 + 630 + 27)$$

$$5F - 160 = 9F - 2457$$

$$2457 - 160 = 4F$$

$$4 \quad 2297 = 4F \quad 574.22$$

$$4 \quad F = \frac{2297}{4} \quad 448.55$$

$$= \frac{42}{574.25} F \quad \neq$$

$$574.25 - 32 = K - 273$$

$$9 \quad (542.25) 5 = 9K - (5 \times 2457)$$

$$9 \quad (542.25) 5 + 2457 = K$$

$$4 \quad K =$$

Soln, speed of light
 $c = 3 \times 10^8 \text{ m/s}$, let that unit be x ,

But, Speed of light = 1 x/s

We know, $v = \frac{d}{t}$

$$c \quad 1 \text{ x/s} = \frac{d}{440}$$

$$d \quad \cancel{440 \text{ x/s}} \neq$$

$$c \quad 440 x$$

Soln, $20 \text{ J} = ?$

c)

→ Given,

$$F = K A \rho V^x$$
$$[M L T^{-2}] = [L^2 M L^{-3} V^x]$$
$$= [M L^{-1} V^x]$$

